



STARNet: Low-light video enhancement using spatio-temporal consistency aggregation

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ABSTRACT

In low-light environments, capturing high-quality videos is an imaging challenge due to the limited number of photons. Previous low-light enhancement approaches usually result in over-smoothed details, temporal flickers, and color deviation. We propose STARNet, an end-to-end video enhancement network that leverages temporal consistency aggregation to address these issues. We introduce a spatio-temporal consistency aggregator, which extracts structures from multiple frames in hidden space to overcome detail corruption and temporal flickers. It parameterizes neighboring frames to extract and align consistent features, and then selectively fuses consistent features to restore clear structures. To further enhance temporal consistency, we develop a local temporal consistency constraint with robustness against the warping error from motion estimation. Furthermore, we employ a normalized low-frequency color constraint to regularize the color as the normal-light condition. Extensive experimental results on real datasets show that the proposed method achieves better detail fidelity, color accuracy, and temporal consistency, outperforming state-of-the-art approaches.

1. Introduction

Low-light environments pose significant challenges for various video applications in visual surveillance, autonomous driving, and astronomy observation [1]. In the extremely low-light conditions, the scarcity of photons results in invisible regions and low signal-to-noise ratios in captured images. While extending exposure time can improve image quality, it is impractical for video applications due to the conflict to high frame rates. Low-light video enhancement (LLVE) techniques aim to generate a bright and clean video from the frames captured under the dim light and backlit conditions, benefiting human visual perception and downstream vision tasks (e.g. detection and segmentation).

Low-light enhancement approaches usually model a transformation from the low-contrast and degraded input to the output with normal brightness, vivid color, and rich details. Previously, LLVE is realized by directly applying the low-light image enhancement (LLIE) techniques on video frames. Traditional low-light enhancement approaches are based on histogram equalization and image decomposition. These approaches lack insights into the inherent degradation process in the extremely low-light conditions, resulting in noisy output.

Recent learning-based works devote to adaptively learning a normal-light luminance distribution. These approaches [2–5] aim to generate clean enhanced images by jointly considering enhancement and denoising. As the low-light enhancement is an ill-posed problem, there are multiple feasible solutions in the solution space. It results the undesired flicker phenomenon when the single-image enhancement approaches are directly applied to videos. To solve the problem, several video-oriented enhancement works [6–8] performs temporal constraint to the siamese network during training. However, when directly applied to enhance dynamic low-light videos, the existing approaches exhibit three drawbacks. First, previous end-to-end works usually concentrate on removing noise but ignore preservation of structures, which results in over-smoothed details. Second, existing temporal losses rely on either static dataset or challenging optical flow warping [6,8], the latter of which struggles with errors of estimation in low-light conditions. Third, the color deviation exists in these works because the color distribution varies between the low-light images and the normal-light environment [9].

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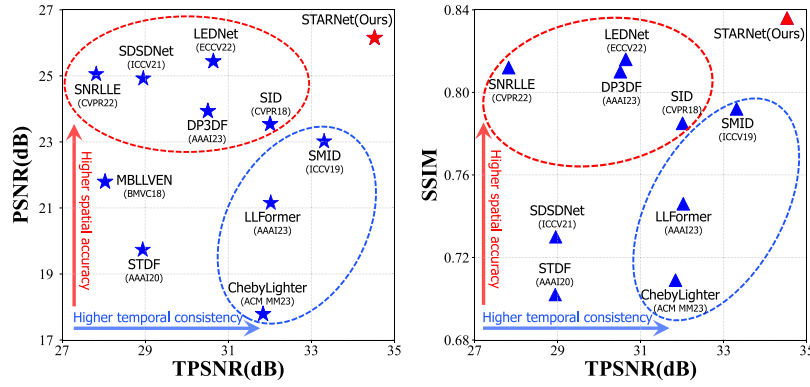


Fig. 1. Spatial and temporal quality comparisons on SDSD dataset [10] in terms of (a) PSNR/TPSNR and (b) SSIM/TPSNR among low-light enhancement approaches, including LLFormer [5], DP3DF [11], ChebyLighter [2], SNRLLE [3], LEDNet [4], SDSDNet [10], STDF [12], SMID [6], SID [13], MBLLVEN [14], and our STARNet. The approaches within the red dashed line offer higher spatial quality at the expense of temporal consistency, whereas those within the blue dashed line prioritize temporal over spatial quality. Our STARNet surpasses these methods in both aspects. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

In this work, we propose a novel end-to-end video enhancement method, namely *STARNet*, to tackle the above three limitations. First, we introduce a spatio-temporal consistency aggregator (STAR) to explicitly use the local spatial information from neighboring frames. Specifically, it uses a motion descriptor to perform consistent structure alignment for multi-frame representations in hidden space, addressing varying pixelwise motions across frames. The aligned consistent feature representations are selectively aggregated to restore sharp edges and details, where the temporal information propagation also preserves the temporal consistency across neighboring frames. It can be applied to other video tasks that benefit from the utilization of multi-frame structural redundancy. Our STAR aggregation can achieve higher temporal consistency without the temporal constraint, outperforming approaches that utilize such constraints. To further produce flicker-free results, we introduce a local temporal consistency constraint into our model. As the statistical characteristics of corresponding patches in neighboring frames are temporally invariant for consecutive frames, we develop a patch-based temporal constraint which is robust to errors of optical flow estimation in existing temporal constraints. To enhance the color accuracy, we develop an intensity-normalized color constraint based on the human visual sensitivity to the low-frequency range [15], which avoids non-uniform supervision of intensity from the uneven illumination occurrence in low-light conditions. With these constraints, our STARNet achieves higher reconstruction accuracy without the extra cost of parameters. As shown in Fig. 1, our method outperforms the state-of-the-art approaches in both spatial quality and temporal consistency, while the others either exhibit lower spatial quality or lack temporal consistency.

Overall, the main contributions of our work are summarized as follows:

- We propose a novel end-to-end low-light video enhancement model that aims to achieve both spatial accuracy and temporal consistency via the consistency aggregation across frames.
- We introduce a spatio-temporal consistency aggregator that extracts and fuses multi-frame consistent features, which preserves more details and enhances temporal consistency.
- We develop effective temporal consistency and color constraints to further refine temporal and spatial quality of enhanced videos.
- Our experimental results on real datasets demonstrate that our STARNet achieves state-of-the-art performance in perspectives of both quantitative evaluations and visual effects, with sharper details, higher color accuracy, and less temporal flickers.

2. Related work

2.1. Low-light image enhancement (LLIE)

Deep learning has achieved success in single-image enhancement methods that early video applications adopted. These algorithms can be categorized into three types: the end-to-end one, the Retinex-based one, and the deep curve-based one. LLNet [16] and LightenNet [17] represent the early deep LLIE models with encoder-decoder architecture. They employ the end-to-end manner to combine enhancement and denoising. Following this idea, recent models are developed with multiple functional components for various sub-tasks [4,18], such as illumination mapping, denoising, or reconstruction. The inherent characteristics of the low-light image are also researched to improve the enhancement quality, such as the distribution of signal-to-noise ratios [3] and the raw format [13]. More recent transformer-based methods like [5] present the capability in LLIE but they also bring more computing cost. Retinex-based methods [19,20] apply the Retinex theory to decompose an image into an illumination component and a reflection component. As Retinex theory enables enhancement models to dispense with supervision, many unsupervised LLIE methods [21,22] are Retinex-based ones as well. Deep curve-based methods like [23] conduct pixelwise Gamma correction iteratively. These methods exclude noise suppression and may amplify the noise signal. In the task of low-light video enhancement, most LLIE methods exhibit an unwanted flicker phenomenon because these models produce unstable results within the solution space when enhancing consecutive frames [24].

2.2. Low-light video enhancement (LLVE)

Compared to LLIE methods, LLVE methods take temporal consistency into consideration to reduce the flicker. The learning-based LLVE methods are obstructed in obtaining paired samples. Since the long exposure conflicts with the high frame rate, capturing the paired frames in both low-light and normal-light conditions is incredibly challenging. SMID [6] presents the first dataset (DRV) for LLVE task, along with an end-to-end LLVE model trained using a pixel-based temporal consistency loss to reduce flickers. However, SMID still suffers shortcomings from the color deviation, and detail over-smoothness. To solve the problems, the dynamic data synthesis approaches [7,8] are developed because DRV only contains static scenes. These synthesis works allow to train models with dynamic data, but the synthesized data lack of real noise distribution. SMOID [25] and SDSD [10] develop real dynamic data with an ND filter and a fixed track respectively. Nevertheless, these methods cannot remove flickers completely. They fail to fully utilize potential of multi-frame information and lack temporal consistency

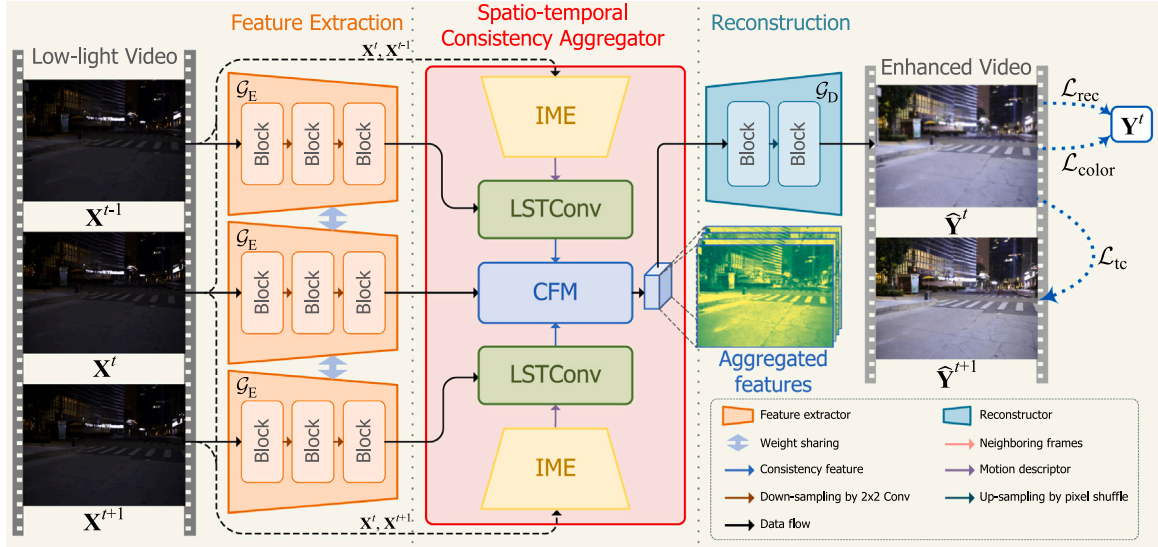


Fig. 2. Architecture of our STARNet. The network consists of three components: a feature extractor ($\mathcal{G}_E$), a spatio-temporal consistency aggregator (STAR), and a reconstructor ($\mathcal{G}_D$). The STAR component extracts and aggregates consistent structures from unaligned features by the inter-frame motion encoder (IME), local spatio-temporal convolution (LSTConv), and consistency fusion module (CFM). Without loss of generality, we show that STARNet uses three frames $\{X^{t-1}, X^t, X^{t+1}\}$ to enhance the reference frame X^t .

supervision for dynamic dataset. Compared with these methods, our framework overcomes the issues by the consistent feature aggregation and spatial and temporal constraints.

3. Method

3.1. STARNet architecture

Compared with conventional approaches, we focus on preserving sharp details and the temporal consistency in low-light video enhancement. The clean video signal is structurally redundant and temporal-consistent, whereas the noise signal is temporal-inconsistent [26]. Therefore, we merge multi-frame to enhance the structures and reduce the variance of noise. As neighboring frames hold consistent structural information to reconstruct the t th frame, we exploit consistent information across neighboring $(2T+1)$ frames to enhance structures and reduce noise simultaneously, which can be formulated by a multi-frame aggregation,

$$Y^t \leftarrow \text{aggregation}(W_{t+\Delta t}^t(X^{t+\Delta t})), \quad \Delta t \in \{-T, \dots, T\} \quad (1)$$

where X^t denotes the t th low-light frame and Y^t denotes the corresponding enhanced result. Specifically, our STARNet extracts consistent features from the neighboring frames $\{X^{t-T}, \dots, X^t, \dots, X^{t+T}\}$, and aggregates multi-frame information in the feature domain to reconstruct the reference frame X^t . Derived from the structural consistency prior, the aggregation also encourages consistency across enhanced frames.

As illustrated in Fig. 2, STARNet consists of a feature extractor, a spatio-temporal consistency aggregator (STAR), and a reconstructor. First, we use the feature extractor $\mathcal{G}_E$ to process the low-light video frame by frame. The feature extractor $\mathcal{G}_E$ maps each low-light frame to a hidden space. Thus, the output features of size $h \times w \times c$ from each frame can be formulated as $z^t = \mathcal{G}_E(X^t)$. Then, to overcome the issues of over-smoothness and noise, we aggregate consistent feature representations from neighboring frames to reconstruct the enhanced reference, corresponding to (1). We perform the STAR component in latent space to achieve this objective. It enables the deep network to aggregate feature representations adaptively, without requiring extra strategy to weight each frame. Our STAR component employs a consistency fusion module (CFM) to selectively fuse the aligned multi-frame features. Denoting the

CFM as $\mathcal{G}_{CFM}$ and the aggregated features as z_A^t , respectively, the STAR component can be expressed as

$$z_A^t = \mathcal{G}_{CFM}(\text{concat}(\{W_{t+\Delta t}^t(z^{t+\Delta t})\}_{\Delta t \in \{-T, \dots, T\}})), \quad (2)$$

where $\text{concat}(\cdot)$ and $W_{t+\Delta t}^t(\cdot)$ denote the concatenation operation and the motion mapping in feature domain, respectively. The reconstructor $\mathcal{G}_D$ maps the aggregated representation back to the image domain, which reconstructs the enhanced result with sharp edges and details.

3.2. Spatio-temporal consistency aggregator

The spatio-temporal consistency aggregator (STAR) aims to extract and fuse the multi-frame information to preserve consistent structures and reduce flickers. It aggregates consistent feature representations through three steps: motion encoding, local alignment, and feature fusion. As shown in Figs. 3 and 5, the STAR consists of an inter-frame motion encoder (IME), a local spatio-temporal convolution (LSTConv), and a consistency fusion module (CFM), where we denote the aligned features from the $(t+\Delta t)$ -th frame to the t th frame as $z_{t+\Delta t}^t$ that $z_{t+\Delta t}^t = W_{t+\Delta t}^t(z^{t+\Delta t})$, $\Delta t \in \{-T, \dots, T\}$. Specifically, the IME module first encodes inter-frame motion and consistent features, and then the LSTConv operation aligns the consistent structural features. Finally, the CFM carries out the feature fusion.

Inter-frame Motion Encoder (IME). Conventional video processing approaches usually use optical flow to model the inter-frame motion [8], but applying accurate optical flow warping is challenging under the weak illumination due to errors of estimation. Instead, our STAR performs alignment and aggregation in the feature domain, allowing us to extract consistent structural features in a learned latent space. The motion extractor is constructed from four residual convolution structures. It extracts a deep descriptor $M_{t+\Delta t}^t \in \mathbb{R}^{h \times w \times (k^2 \times c)}$ for the motion from $X^{t+\Delta t}$ to X^t , formulated as $M_{t+\Delta t}^t = \mathcal{G}_{IME}(X^{t+\Delta t}, X^t)$. The deep motion descriptor models the dynamic local motion mapping in the feature domain.

Local Spatio-temporal Convolution (LSTConv). As the inter-frame motion varies among pixels, the alignment conducts a spatio-temporal translation-variant processing. Inspired by [27], the LSTConv operation parameterizes the motion descriptor from the IME module to align the consistent feature representations across neighboring frames with the reference frame. First, the local motion descriptor at the coordinates (x, y) is reshaped into a local mapping kernel $K_{x,y} \in \mathbb{R}^{k \times k \times c}$,

$$K_{x,y} = \text{reshape}_{k \times k} (M_{t+\Delta t}^t(x, y)). \quad (3)$$

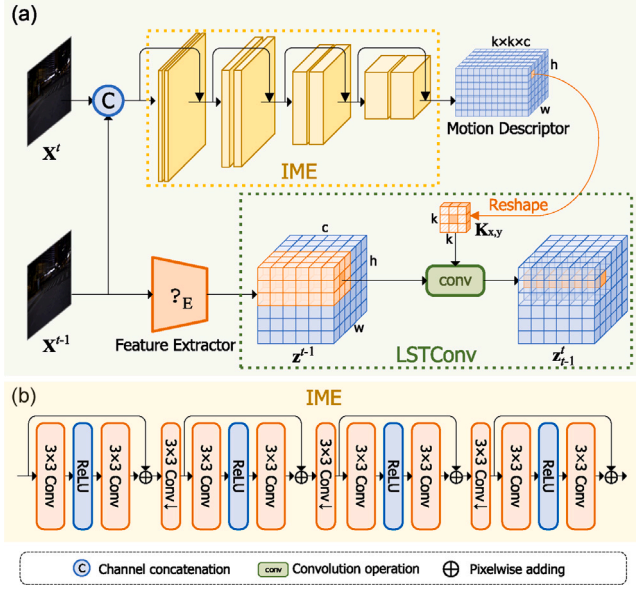


Fig. 3. Illustration of the inter-frame motion encoder (IME) and the local spatio-temporal convolution (LSTConv) in the spatio-temporal consistency aggregator (STAR). (a) The IME module and LSTConv operation that extract consistent structure from the $(t-1)$ th-frame. (b) The internal structures of the IME module.

We set $k = 5$ in this work. Then, we perform a local convolution between the local mapping kernel $K_{x,y}$ and the corresponding unaligned local features $z^{t+dt}(x, y)$. The local convolution to generate the aligned features is formulated as

$$z_{t+dt}^t(x, y) = \sum_{u=-\lfloor \frac{k}{2} \rfloor}^{\lfloor \frac{k}{2} \rfloor} \sum_{v=-\lfloor \frac{k}{2} \rfloor}^{\lfloor \frac{k}{2} \rfloor} K_{x,y}(u, v) z^{t+dt}(x-u, y-v), \quad (4)$$

As the signal-to-noise ratio of structural signal is extremely low under low-light conditions, differing from [27], LSTConv operation utilizes deep representations to effectively model inter-frame motions of weak structures in both preceding and succeeding frames. Fig. 4 visualizes features in the STAR component, which shows that the LSTConv operation extracts the consistent structure and removes the inter-frame motion.

Consistency Fusion Module (CFM). The CFM aims to selectively fuse features. Due to the occlusion and severe degradation, the consistent information in each frame (either the low-light reference frame or the neighboring frame) is usually inadequate to reconstruct the enhanced frame. Hence, we selectively fuse these multi-frame features to obtain abundant information. As a direct fusion of the low-resolution features may lose sub-pixel information, we apply a pixel shuffle operation that utilizes subpixel information to prevent spatial structural degradation. Considering that the amount of information contained in each frame is not evenly distributed, we concatenate these multi-frame features in channel dimension, and apply a channel attention block to adaptively weight each channel. The CFM outputs the aggregated feature representation z_A^t . By selectively fusing consistent features, the aggregated features are enriched with abundant structural information, facilitating high-quality reconstruction.

3.3. Loss function

Our model is trained with three losses: the reconstruction loss, the local temporal consistency loss, and the normalized low-frequency color loss. First, we exploit the reconstruction loss to restore the content for each frame. The reconstruction loss comprises the l_1 -loss and the perceptual loss,

$$\mathcal{L}_{\text{rec}} = \|\mathbf{Y}^t - \hat{\mathbf{Y}}^t\|_1 + \mathcal{L}_{\text{per}}, \quad (5)$$

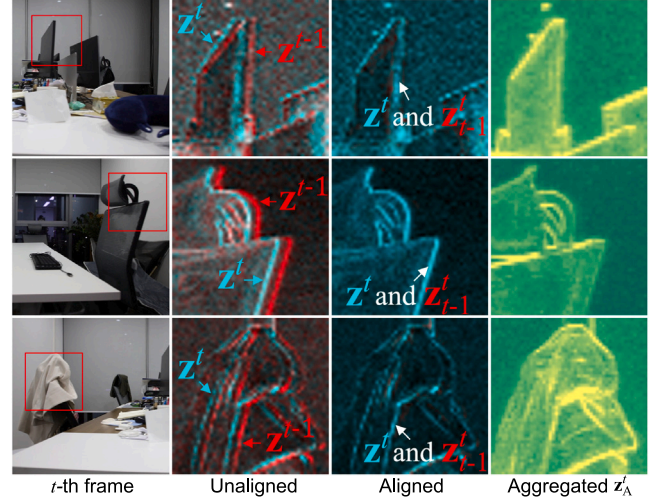


Fig. 4. Visualization of extraction, alignment, and fusion of consistent features in the STAR component. The inter-frame motion encoder (IME) and local spatio-temporal convolution (LSTConv) aligns the features z^{t-1} to z^t . The z_{t-1}^t coincides with z^t after alignment.

where $\mathbf{Y}^t$ denotes the ground truth for the t th frame. $\mathcal{L}_{\text{per}}$ denotes the perceptual loss in [28], which is constructed based on the pretrained VGG-19.

Local temporal consistency constraint. Conventional temporal consistency losses rely on accurate optical flow warping [8], which is sensitive to estimation error. To achieve a better temporal consistency, we design a local temporal consistency loss based on statistical characteristics consistency of patches. As the inter-frame motion in a local patch is usually finite, the statistical characteristics of the corresponding patches in the two frames should be consistent. Hence, we constrain the statistical consistency of corresponding patches between two neighboring enhanced frames. Specifically, we crop each frame into N_p non-overlapping $w \times w$ patches $\hat{\mathbf{p}} \in \mathbb{R}^{w \times w \times 3}$. Denoting the mean of the j th patch $\hat{\mathbf{p}}_j$ in the enhanced t th frame as $\bar{p}_j^t$, the local temporal consistency loss is expressed as

$$\mathcal{L}_{\text{tc}} = \frac{1}{N_p} \sum_{j=1}^{N_p} \|\bar{p}_j^t - \bar{p}_j^{t+1}\|_1, \quad (6)$$

where $\bar{p}_j^t = \text{mean}(\mathcal{W}_{t+1}^t(\hat{\mathbf{p}}_j^{t+1}))$. The optical flow warping $\mathcal{W}_{t+1}^t(\cdot)$ only coarsely aligns patches to handle the case of extremely large shift. Owing to discarding the pixelwise regularization, the patch-based constraint is robust to errors in pixelwise motion estimation. In our implementation, we apply the optical flow in [29] and set $w = 64$.

Normalized low-frequency color constraint. To correct the color deviation, we consider the frequency characteristic of vision perception and practical uneven illumination. First, as the chromatic perception in human vision system is sensitive to the low-frequency band [15], our color supervision is based on the regularization of the low-frequency component. Then, as the low-light environment usually accompanies with the uneven intensity distribution, e.g. several small bright regions surrounded by a flat dark background. The local bright region with higher absolute intensity may dominate over the supervision from flat extremely low-light regions, which should be avoided. Combining the two points, we constrain the channel-wise normalized low-frequency component for higher color accuracy. Our color loss is expressed as

$$\mathcal{L}_{\text{color}} = \sum_{i \in \{R, G, B\}} \|\mathcal{N}_i(\text{LPF}(\mathbf{Y}^t)) - \mathcal{N}_i(\text{LPF}(\hat{\mathbf{Y}}^t))\|_1, \quad (7)$$

where $\text{LPF}(\cdot)$ denotes the 3×3 Gaussian low-pass filtering. The normalized operator $\mathcal{N}(\cdot)$ denotes computing normalized RGB ratio, $\mathcal{N}_i(\mathbf{Y}) = \mathbf{Y}_i \oslash (\mathbf{Y}_R + \mathbf{Y}_G + \mathbf{Y}_B)$. Here $\mathbf{Y}_R$, $\mathbf{Y}_G$, and $\mathbf{Y}_B$ denote the red, green and blue channel of the enhanced frame $\mathbf{Y}$, respectively.

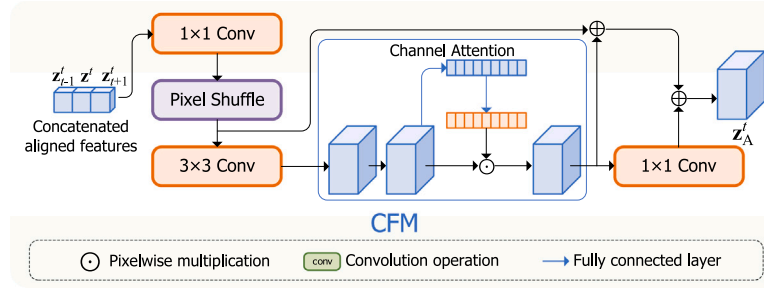


Fig. 5. Illustration of the consistency fusion module (CFM). The CFM selectively fuses multi-frame structure information from the concatenation of consistent structural features and outputs the aggregated features z_A^l .

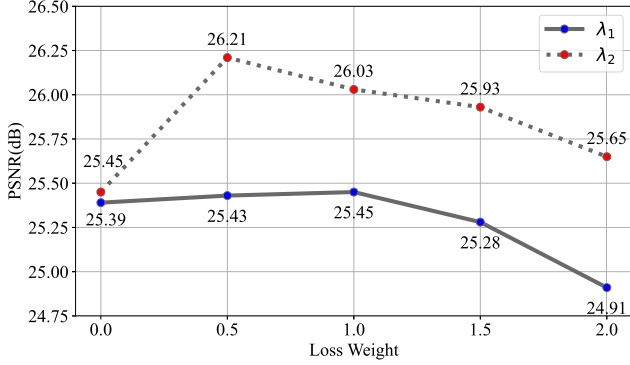


Fig. 6. Hyper-parameter tuning in the total loss.

Total loss. The total loss is expressed as

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda_1 \mathcal{L}_{\text{tc}} + \lambda_2 \mathcal{L}_{\text{color}}, \quad (8)$$

where λ_1 and λ_2 balance the local temporal consistency loss and the color loss, respectively.

3.4. Implementation details

We set $T = 1$ in our implementation and apply the NAFBlock [30] as the basic block to build the feature extractor and reconstructor. We implement our model with PyTorch platform. The model is trained with the siamese scheme as [6]. Since low-light video is processed in a frame-by-frame scheme, the feature representation for each frame can be reused in enhancing the subsequent frames, which decreases the computation workload. Each training sample is randomly cropped into 512×512 patches. We apply the Adam optimizer of $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The initial learning rate is set to 1×10^{-4} , decaying by the cosine-annealing scheme. As shown in Fig. 6, we perform a grid search to find the suitable hyper-parameters. We search λ_1 by setting $\lambda_2 = 0$, and then, fix the promising $\lambda_1 = 1.0$ to search λ_2 . Based on the result of grid search, we set $\lambda_1 = 1.0$ and $\lambda_2 = 0.5$, respectively.

4. Experiments

4.1. Experimental setup

We conduct both quantitative and qualitative evaluations in the experiments. We compare our STARNet with five state-of-the-art LLIE approaches (ChebyLighter [2], SID [13], LLFormer [5], SNRLLE [3], and LEDNet [4]) and five LLVE approaches (MBLLVEN [14], SMID [6], STDF [12], SDSDNet [10], and DP3DF [11]). We perform the comparison experiments on an NVIDIA RTX8000 GPU. To evaluate our STARNet on real data, we use the public SDSD [10] and DRV [6] datasets in the experiments. The DRV dataset is a static LLVE dataset,

containing 202 indoor and outdoor scenes, while the SDSD dataset is the first real dynamic LLVE dataset that contains 150 scenes. We randomly divide 1/3 of total data for test, regardless of whether the scenes are indoor or outdoor, and the others for training. Our model is trained with 2000 and 1000 epochs on SDSD and DRV dataset, respectively. We retrain all the compared methods with the same training data as ours for fairness.¹ We evaluate the approaches on all test samples, whose image resolution is 1920×1080 for the SDSD dataset and 1374×918 for the DRV dataset.

4.2. Quantitative evaluation

We use the peak signal-to-noise ratio (PSNR), structural similarity (SSIM), learned perceptual image patch similarity (LPIPS), mean absolute error (MAE), and full-structure similarity index (FSIM) to analyze the image quality of the enhancement results. For the real dynamic SDSD dataset, we randomly select 10 continuous frames from each scene and then evaluate the average metrics of enhanced results. For the static DRV dataset, we adopt the fifth-frame evaluation scheme as SMID [6]. The quantitative evaluation results for SDSD and DRV are presented in Table 1 and Table 2, respectively. Our STARNet achieves the best performance in generating high-fidelity enhanced frames close to the normal-light capture. Moreover, we apply TPSNR, TSSIM, and TMAE as [6] to evaluate the temporal consistency across frames. These temporal metrics can be expressed as

$$\text{TPSNR}(\hat{\mathbf{Y}}^t, \hat{\mathbf{Y}}^{t+1}) = \text{PSNR}(\hat{\mathbf{Y}}^t - \mathcal{W}_{t+1}^t(\hat{\mathbf{Y}}^{t+1})), \quad (9)$$

$$\text{TSSIM}(\hat{\mathbf{Y}}^t, \hat{\mathbf{Y}}^{t+1}) = \text{SSIM}(\hat{\mathbf{Y}}^t - \mathcal{W}_{t+1}^t(\hat{\mathbf{Y}}^{t+1})), \quad (10)$$

$$\text{TMAE}(\hat{\mathbf{Y}}^t, \hat{\mathbf{Y}}^{t+1}) = \text{MAE}(\hat{\mathbf{Y}}^t - \mathcal{W}_{t+1}^t(\hat{\mathbf{Y}}^{t+1})), \quad (11)$$

where the warping function $\mathcal{W}_{t+1}^t(\cdot)$ is computed by optical flow estimation. We perform the comparison of temporal consistency on the real dynamic SDSD dataset. As shown in Table 1, our STARNet achieves the best temporal consistency. To further validate the effectiveness of our multi-frame aggregation, we also compare the enhanced results of “LLIE network+flow” strategy with our method. Specifically, we apply the low-light optical flow warping [31] to two state-of-the-art LLIE methods (SID [13] and LEDNet [4]) so that the input of their networks is the concatenation of aligned three frames. The comparisons in Table 1 show our consistency aggregation method achieves better spatial and temporal performance than the image-domain aggregation strategy.

Lightness accuracy. To demonstrate the accuracy of lightness, we investigate the mean absolute error of lightness between the enhanced result and the ground truth. Here, we apply the definition of lightness as the CIE Lab color space. Table 3 shows that our STARNet achieves the highest accuracy of lightness among the state-of-the-art methods. We visualize the mean absolute error of lightness in Fig. 7. As the

¹ We use the same random seed of 1.

Table 1

Quantitative results on the SDSD dataset, in terms of spatial metrics (PSNR(dB), SSIM, LPIPS, MAE, and FSIM) and temporal metrics (TPSNR(dB), TSSIM, and TMAE). Results of SDSDNet are cited from [10].

Method	PSNR↑	SSIM↑	LPIPS↓	MAE↓	FSIM↑	TPSNR↑	TSSIM↑	TMAE _(0.01) ↓
ChebyLighter [2]	17.787	0.709	0.327	0.116	0.879	31.839	0.875	1.625
SID [13]	23.539	0.785	0.302	0.056	0.894	32.009	0.927	1.790
LLFormer [5]	21.154	0.746	0.363	0.072	0.891	32.025	0.918	1.659
SNRLLE [3]	25.052	0.812	0.240	0.045	0.916	27.822	0.901	3.593
LEDNet [4]	25.442	0.816	0.230	0.044	0.915	30.642	0.934	1.606
SID+flow	24.030	0.797	0.302	0.055	0.896	31.023	0.942	1.466
LEDNet+flow	25.455	0.815	0.231	0.042	0.911	32.533	0.944	1.438
MBLLVEN [14]	21.794	0.652	0.271	0.074	0.892	28.036	0.869	3.075
SMID [6]	23.015	0.792	0.278	0.062	0.896	33.309	0.947	1.339
STDF [12]	19.733	0.702	0.402	0.085	0.834	28.943	0.877	2.531
SDSDNet [10]	24.920	0.730	–	–	–	28.955	0.774	2.467
DP3DF [11]	23.933	0.810	0.301	0.063	0.912	30.511	0.920	1.601
STARNet (Ours)	26.210	0.839	0.208	0.040	0.935	34.524	0.951	1.005

Table 2

Quantitative results on the DRV dataset, in terms of PSNR(dB), SSIM, LPIPS, MAE, and FSIM. Results of SDSDNet are cited from [10].

Method	PSNR↑	SSIM↑	LPIPS↓	MAE↓	FSIM↑
ChebyLighter [2]	23.720	0.077	0.340	0.055	0.946
SID [13]	27.592	0.802	0.311	0.040	0.959
LLFormer [5]	27.767	0.789	0.308	0.038	0.962
SNRLLE [3]	28.306	0.836	0.244	0.031	0.971
LEDNet [4]	28.950	0.843	0.236	0.029	0.972
MBLLVEN [14]	26.885	0.803	0.316	0.047	0.953
SMID [6]	28.328	0.825	0.256	0.032	0.961
STDF [12]	25.034	0.795	0.313	0.045	0.956
SDSDNet [10]	26.030	0.750	–	–	–
DP3DF [11]	26.107	0.813	0.306	0.042	0.960
STARNet (Ours)	29.904	0.872	0.198	0.025	0.983

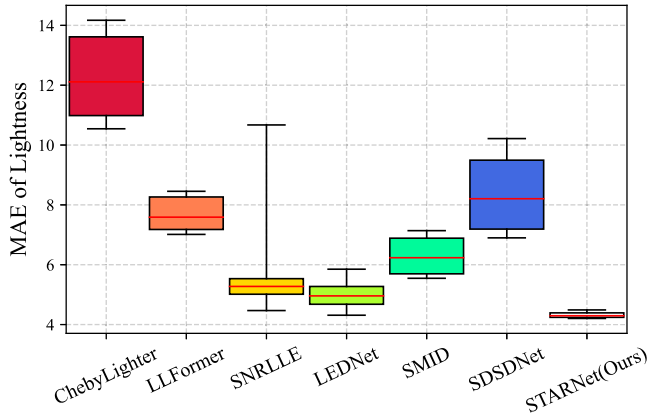


Fig. 7. Mean absolute error of lightness on tested SDSD videos. Our STARNet achieves the lowest lightness error and the minimal fluctuation among the state-of-the-art methods.

ground truth is ideally temporal consistent, our method exhibits a closer adherence to the temporal consistency of lightness compared to other state-of-the-art approaches.

Color accuracy. To illustrate the color accuracy, we analyze the ΔE_{ab} [32] between the enhanced results and the ground truth, where the smaller ΔE_{ab} value reveals the less color deviation. The results in Table 3 indicate that our STARNet achieves the highest color accuracy. Compared with other state-of-the-art approaches, our color constraint is capable of generating videos with vivid color, close to the visual effect under the normal-light condition.

Detail restoration. We apply multi-scale structural similarity (MS-SSIM) [33] and deep image structure and texture similarity (DISTS)

Table 3

Comparison of the lightness accuracy and the color deviation on the SDSD and DRV datasets, in terms of the mean absolute error of lightness and the ΔE_{ab} , respectively.

Method	MAE of Lightness↓		ΔE_{ab} ↓	
	SDSD	DRV	SDSD	DRV
ChebyLighter [2]	12.142	4.366	14.665	13.043
LLFormer [5]	7.584	2.862	10.284	8.747
SNRLLE [3]	5.188	2.314	6.070	7.406
LEDNet [4]	4.754	2.182	5.841	6.820
SMID [6]	6.280	2.385	8.227	7.677
SDSDNet [10]	8.152	4.103	11.422	14.238
DP3DF [11]	6.371	3.805	8.027	9.619
STARNet (Ours)	4.350	2.038	5.504	6.597

Table 4

Comparison of detail restoration on the SDSD dataset, in terms of MS-SSIM and DISTS.

Method	MS-SSIM↑	DISTS↓
ChebyLighter [2]	0.822	0.169
LLFormer [5]	0.847	0.208
SNRLLE [3]	0.850	0.107
LEDNet [4]	0.889	0.105
SMID [6]	0.871	0.174
SDSDNet [10]	0.769	0.199
DP3DF [11]	0.894	0.171
STARNet (Ours)	0.910	0.101

[34] to compare the performance of detail restoration. MS-SSIM introduces multi-scale analysis to comprehensively consider the detail information of images. DISTS measures the distance between the multi-scale deep representations to evaluate the structural and textural similarity. Table 4 shows that our consistency aggregation achieves higher fidelity in detail restoration than other state-of-the-art approaches.

4.3. Qualitative evaluation

We compare our STARNet with other state-of-the-art approaches by qualitative analysis on the SDSD dataset (Fig. 8 and Fig. 9) and the DRV dataset (Fig. 10 and Fig. 11), respectively. Our STARNet achieves better visual effects in both indoor and outdoor scenes. Benefiting from our STAR aggregation and constraints, the STARNet can obtain bright enhanced results with less noise and fewer artifacts than ChebyLighter and LLFormer. Compared with DP3DF, SMID, and LEDNet, our STARNet preserves sharper details. With effective aggregation and the color constraint, our STARNet is capable of generating enhanced frames with higher color accuracy of flat regions than ChebyLighter, SNRLLE, SDSDNet, and SMID.

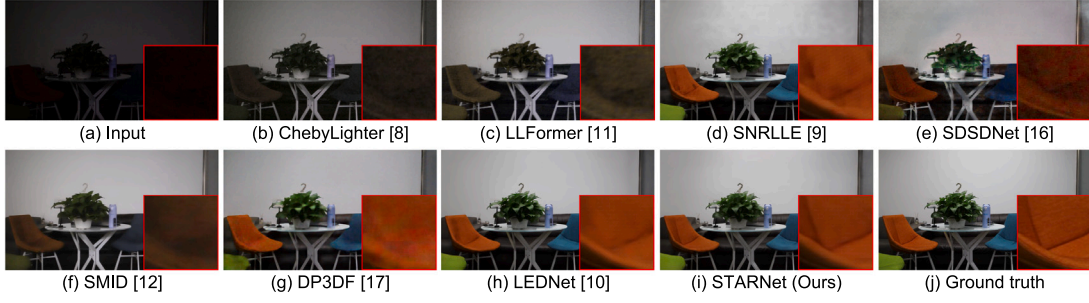


Fig. 8. Enhanced results produced by our STARNet and other eight state-of-the-art approaches on the SDSD indoor scene.

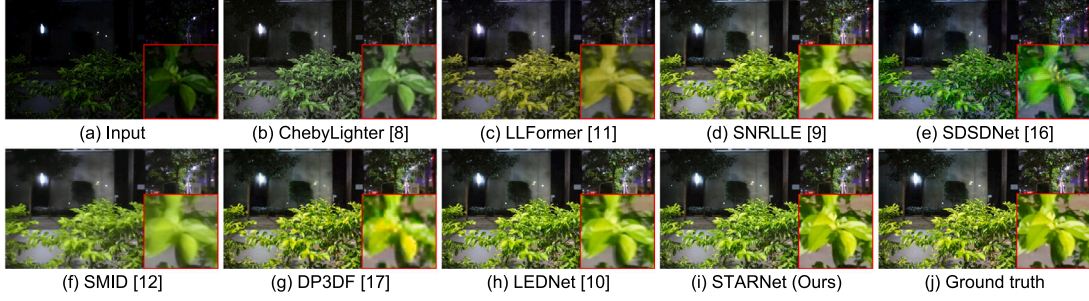


Fig. 9. Enhanced results produced by our STARNet and other eight state-of-the-art approaches on the SDSD outdoor scene.

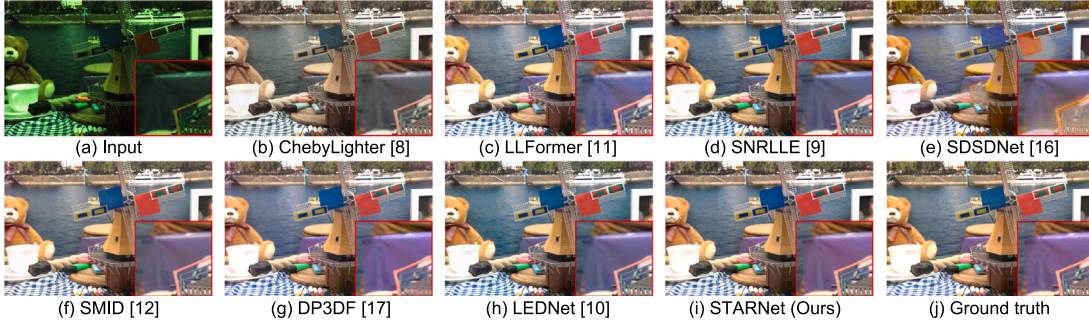


Fig. 10. Enhanced results produced by our STARNet and other eight state-of-the-art approaches on the DRV indoor scene.

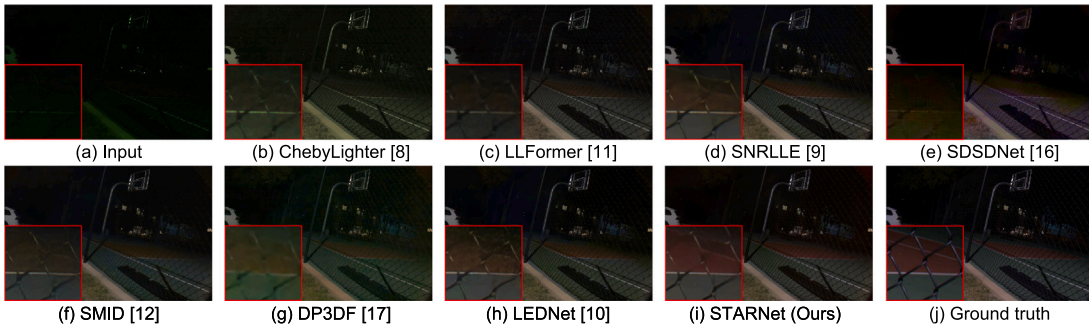


Fig. 11. Enhanced results produced by our STARNet and other eight state-of-the-art approaches on the DRV outdoor scene.

4.4. Ablation study

We perform ablation study to evaluate the effectiveness of the modules of our STARNet by five experimental settings: (1) replacing the multi-frame input with 1-frame input to evaluate the effectiveness of multi-frame aggregation idea, (2) replacing the STAR aggregation with the backbone blocks of the same parameter size to evaluate the effectiveness of STAR, (3) keeping the setting (2) and applying optical flow alignment at the input, (4) replacing the CFM with the backbone

blocks of the same parameters size to evaluate effectiveness of the CFM, (5) removing our local temporal consistency loss, (6) removing our normalized low-frequency color loss, and (7) full model.

Table 5 lists the spatial and temporal evaluation results. We can find that the multi-frame information benefits the upgrade in spatial quality, as evidenced by the 1-frame setting achieving the worst results. By comparing results of the setting (2), (3), and (7), we also validate that our STAR component boosts the multi-frame structural aggregation. Additionally, the temporal comparison of setting (2), (3), and

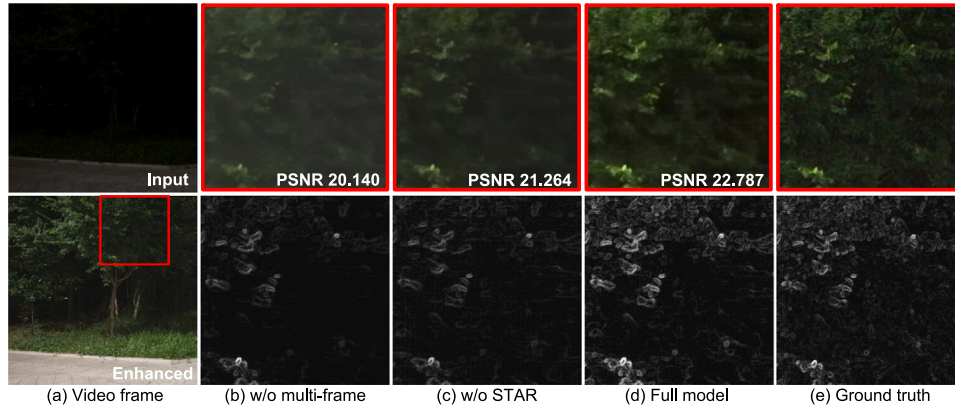


Fig. 12. Enhanced details and corresponding edge detection results produced by three ablation settings about detail preservation. (a) Input and enhanced frame. (b) Without multi-frame. (c) Without STAR feature aggregation. (d) Full model. (e) Ground truth.

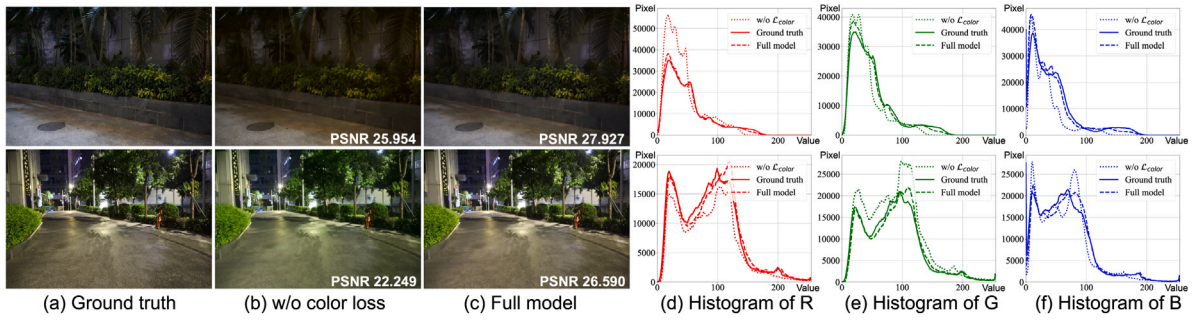


Fig. 13. Visual comparison of enhanced results and corresponding histograms produced by the ablation setting about color accuracy. (a) Ground truth. (b) Without our color loss. (c) Full model. (d) Histogram of red channel. (e) Histogram of green channel. (f) Histogram of blue channel. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 5

Ablation results on the SDS dataset, in terms of PSNR(dB), SSIM, TPSNR(dB), and TSSIM.

Setting	PSNR↑	SSIM↑	TPSNR↑	TSSIM↑
(1) w/o multi-frame	24.231	0.808	31.879	0.938
(2) w/o STAR	25.302	0.809	33.096	0.943
(3) w/o STAR + w/ flow	25.273	0.815	33.588	0.945
(4) w/o CFM	25.051	0.820	33.559	0.924
(5) w/o $\mathcal{L}_{tc}$	—	—	33.950	0.949
(6) w/o $\mathcal{L}_{color}$	25.453	0.825	—	—
(7) Full model	26.210	0.839	34.524	0.951

(7) presents the effectiveness of our consistency aggregation to flicker suppression, as the aggregation is naturally based on the consistency prior of video. Our consistency aggregation achieves higher temporal consistency even without an explicit temporal loss than compared state-of-the-art methods. Besides, the result of setting (4) shows that the CFM effectively fuses consistent information across multi-frame features, producing better detail reconstruction.

Fig. 12 presents the edge detection result by the Sobel operator, which shows the our STAR consistency aggregation is effective to preserve sharp detail edges of leaves. Fig. 13 visualize the two visual comparison examples and corresponding histograms of R, G, and B channels, which demonstrates that our color supervision effectively corrects the color deviation, facilitating higher color accuracy.

4.5. Generalization to challenging scenarios

Low frame rate. Considering that the frame rate is limited by the camera hardware, a lower frame rate may result in a larger inter-frame

Table 6

Generalization performance of our STARNet for lower frame rates, in terms of PSNR(dB), SSIM, TPSNR(dB), and TSSIM.

Setting	PSNR↑	SSIM↑	TPSNR↑	TSSIM↑
w/ 100% frame rate	26.210	0.839	34.524	0.951
w/ 50% frame rate	25.682	0.827	30.861	0.933
w/ 25% frame rate	25.325	0.819	30.317	0.902

shift, which reduces the structural redundancy. We analyze the generalization performance of our STARNet for lower frame rates on the SDS dataset. We train our model with the 100% frame rate and conduct inference at 100%, 50%, and 25% frame rates, respectively, as shown in Table 6. Comparing to other state-of-the-art approaches (e.g. SNRLLE and SDSNet) of 100% frame rate in Table 1, our STARNet achieves comparable in both spatial and temporal quality even with challenging lower frame rates, which shows the effectiveness of our STARNet in large shift cases.

High resolution. We perform additional generalization experiments on the SDS dataset to validate the model effectiveness on higher resolution videos. We apply the bilinear interpolation to simulate $\times 1.5$ and $\times 2$ resolutions, where the $\times 2$ resolution (3840×2160) is regarded as the 4K video in the ultra-high-definition standard. We compare our STARNet with two competing approaches. As shown in Table 7, our STARNet achieves the best spatial accuracy and temporal consistency when generalizing to higher resolutions.

Illumination. To validate the generalizability in various low-light conditions, we perform the Gamma correction, $\mathbf{X}' = \mathbf{X}^\gamma$, on the test samples to obtain the low-light input $\mathbf{X}'$ with various light levels, where $\gamma \in \{0.6, 0.8, 1.0, 1.5, 2.0\}$. We use two competing approaches, LEDNet [4]

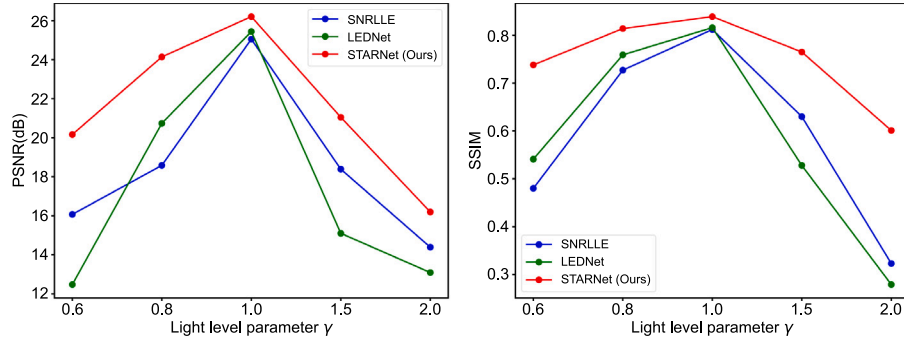


Fig. 14. PSNR and SSIM values of SNRLLE [3], LEDNet [4], and our STARNet under different low-light conditions.

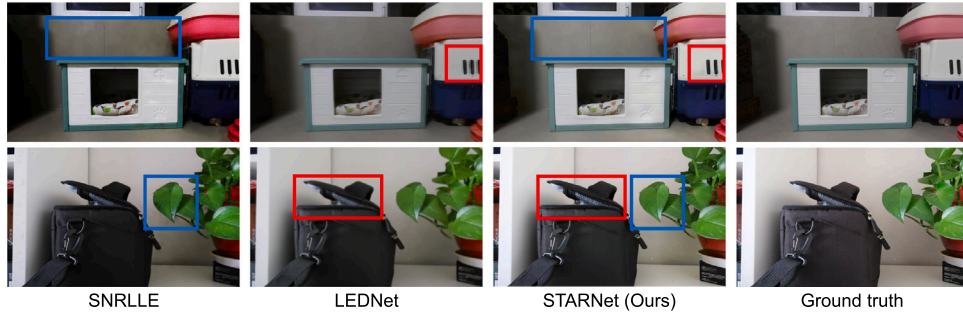


Fig. 15. Generalized results produced by our STARNet and two competing approaches on the DID dataset.

Table 7

Comparison of SNRLLE [3], LEDNet [4], and our STARNet on higher resolution videos in terms of PSNR(dB), SSIM, TPSNR(dB), and TSSIM.

Resolution	Method	PSNR $\uparrow$	SSIM $\uparrow$	TPSNR $\uparrow$	TSSIM $\uparrow$
$\times 1.5$	SNRLLE [3]	24.649	0.823	26.632	0.897
	LEDNet [4]	24.964	0.831	30.401	0.930
	STARNet (Ours)	25.116	0.840	33.835	0.942
$\times 2$	SNRLLE [3]	23.241	0.821	26.450	0.901
	LEDNet [4]	22.901	0.819	29.201	0.835
	STARNet (Ours)	24.205	0.829	33.754	0.948

and SNRLLE [3], for the comparison. As shown in Fig. 14, our STARNet achieves the best performance under various low-light conditions.

Other scene. To validate the generalization to other dynamic scenes, we compare our model with two competing approaches, LEDNet and SNRLLE, on the external DID dataset [35]. Compared with other state-of-the-art methods, our result has clear details and fewer artifacts, as shown in Fig. 15.

4.6. Efficiency

We analyze the model efficiency in perspectives of the parameter size, computation cost, inference time, and inference GPU memory. Table 8 presents the efficiency metrics of our method and other state-of-the-art approaches during processing a 256×256 patch. We also report the efficiency for each component of our STARNet. Our STARNet keeps the similar model size, computational cost, inference time, and memory usage to other approaches but achieves better performance.

4.7. Application to other restoration task

To evaluate the effectiveness of our STAR aggregation on other restoration tasks, we perform experiments on video denoising. We train

the model at $T = 1$ on DAVIS² training dataset with the additive white Gaussian noise of $\sigma \in [5, 50]$. We compare our STARNet with five state-of-the-art video denoising approaches (FastDVDNet-RCN [36], FastDVDNet [37], DVDNet [38], VNLNet [39], and DnCNN [40]) on DAVIS testing dataset. Table 9 shows the average PSNR of our method and other state-of-the-art denoising approaches, where results of DnCNN and VNLNet are cited from [39]. Our STARNet achieves higher PSNR in than other approaches. Since the integration of consistent information restores the degraded content, our consistency aggregation effectively improve the fidelity of reconstruction, demonstrating the potential for application to various multi-frame restoration tasks.

5. Conclusions

In this work, we have proposed the STARNet to address three main problems in the LLVE task: detail over-smoothness, temporal flickers, and color deviation. Compared with previous approaches based on single-image enhancement, our method leverages the multi-frame structural redundancy of the video signal to avoid over-smoothness and suppress noise. We design the STAR component to aggregate the multi-frame consistent structures in the latent space, and introduce the temporal local consistency constraint to assist with flicker suppression. For the color deviation issue, we develop the normalized low-frequency color constraint to correct chromatic errors in results. Extensive experiments demonstrate that our method achieves the state-of-the-art performance on two challenging benchmarks.

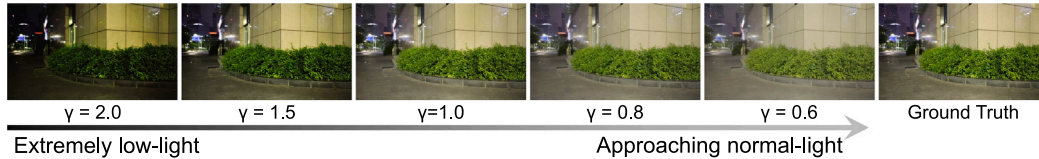
Limitations. Although our STARNet performs better than existing approaches, it still has the limitation in generalizing to the extremely low-light cases. For example, in Fig. 16, our method produces the suboptimal visibility for dark regions in the case $\gamma = 2.0$. This is because the deep learning models are usually constrained by the distribution range of the training data. Applying the stronger data augmentation

² The 2017 DAVIS challenge on video object segmentation.

Table 8

Model size, computational consumption, inference time, and GPU memory of our STARNet and other state-of-the-art approaches.

Method/Module	Parameters(M)	MACs(G)	Inference time(ms)	GPU memory(GB)
ChebyLighter [2]	0.178	11.470	8.09	0.532
LLFormer [5]	24.518	22.035	107.12	9.054
SNRLLE [3]	39.124	23.958	8.62	0.569
LEDNet [4]	7.405	192.858	12.10	1.260
SMID [6]	85.635	33.714	30.80	0.854
SDSDNet [10]	4.299	45.747	29.77	0.756
STARNet-STAR module	4.575	19.630	5.19	0.405
STARNet-feature extractor	3.896	12.139	3.11	0.576
STARNet-reconstructor	0.397	2.275	2.50	0.164
STARNet-full model (Ours)	8.868	34.044	10.82	1.145

**Fig. 16.** Generalized results produced by our STARNet and two competing approaches on the DID dataset.**Table 9**

Average PSNR(dB) of the denoising results of additive Gaussian white noise on the DAVIS testing dataset.

Method	$\sigma = 10$	$\sigma = 20$	$\sigma = 30$	$\sigma = 50$
DnCNN [40]	36.80	32.94	–	–
VNLNet [39]	39.07	35.46	–	–
DVDNet [38]	35.72	34.08	31.40	31.18
FastDVDNet [37]	38.68	35.74	31.58	31.39
FastDVDNet-RCD [36]	38.81	35.69	34.04	31.86
STARNet (Ours)	40.28	36.56	34.21	31.89

or fine-tuning the model for specific scenes will alleviate this issue but require additional training time or data. Our future work will focus on further optimizing the visual effect for the out-of-domain conditions.

CRedit authorship contribution statement

Zhe Wu: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Zehua Sheng:** Writing – review & editing, Methodology, Conceptualization. **Xue Zhang:** Writing – review & editing, Visualization. **Si-Yuan Cao:** Writing – review & editing, Visualization, Methodology. **Runmin Zhang:** Writing – review & editing, Visualization. **Beinan Yu:** Writing – review & editing, Data curation. **Chenghao Zhang:** Writing – original draft, Data curation. **Bailin Yang:** Writing – review & editing, Funding acquisition. **Hui-Liang Shen:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.patcog.2024.111180>.

Data availability

Data will be made available on request.

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